Functions:

- Niacin is found in the form of its amide which is an important constituent of the coenzymes NAD^+ and $NADP^+$ and they take part in redox reactions which are associated with many dehydrogenases. For example NAD^+ is the coenzyme for lacate dehydrogenase. $NADP^+$ is the coenzyme for glutathione reductase.
- It promotes the formation of fat from carbohydrates.
- Niacin is essential for healthy skin, normal functions of the gastro intestinal tract and maintenance of the nervous system.

Absorption and storage

Nicotinic acid and nicotinamide are absorbed by the intestine through the portal vein into the general circulation. Excess nicotinic acid is not stored in the body.

Deficiency:

- Nicotinic acid deficiency causes a disease called pellagra (Figure 8.20).
- Dermatitis-skin exposed to the sun, soreness of the mouth and swelling of the tongue.
- Diarrhoea.
- Dementia-mental changes including depression and confusion.



Figure 8.20 Deficiency due to Vitamin B_3 (Pellagra)

iv. Vitamin B_6 (Pyridoxine)

Vitamin B_6 or Pyridoxine is also called as adermin. Vitamin B_6 consists of three closely related pyridine derivatives.

- Pyridoxine
- Pyridoxal
- Pyridoxal amine

The metabolically active form of vitamin B_6 is pyridoxal phosphate. It is formed from pyridoxal.

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